Immunology and Inflammation

Class A Scavenger Receptor MARCO negatively regulates Ace expression and aldosterone production

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eLife Assessment

O'Brien and co-authors addressed how statins reduce levels of aldosterone in humans and provide **important** data demonstrating that tissue-resident macrophages can exert physiological functions and influence endocrine systems. However, the strength of evidence, as of now, is **incomplete**, as the sole description of the phenotype of MARCO-deficient mice is insufficient to claim that MARCO in alveolar macrophages can negatively regulate ACE expression and aldosterone production at steady-state. The work will be of broad interest to cell biologists and immunologists.

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Abstract

Aldosterone is a potent cholesterol-derived steroid hormone that plays a major role in controlling blood pressure via regulation of blood volume. The release of aldosterone is typically controlled by the renin-angiotensin aldosterone system, situated in the adrenal glands, kidneys, and lungs. Here, we reveal that the class A scavenger receptor MARCO, expressed on alveolar macrophages, negatively regulates aldosterone production and suppresses angiotensin converting enzyme (Ace) expression in the lungs of male mice. Collectively, our findings point to alveolar macrophages as additional players in the renin-angiotensin-aldosterone system and introduce a novel example of interplay between the immune and endocrine systems.

Introduction

The scavenger receptor MARCO (macrophage receptor with collagenous structure), a class A scavenger receptor, is primarily expressed on macrophages including tumour-associated macrophages, and lung macrophages[1,2]. MARCO, upregulated in response to pathogenic challenge[2], has been implicated in defence against pathogens in the lung[3], phagocytosis and clearance of tumor cells [4], internalisation of exosomes [5], and has been touted as a potential target in anti-cancer immunotherapy [6]. The class A scavenger receptors have a broad ligand specificity and similar domain structures, comprising a cytoplasmic tail, transmembrane region, spacer region, α helical coiled domain, collagenous domain, and a C-terminal cysteine-rich domain [7]. Known endogenous ligands for the receptor include, like other class A receptors, modified forms of low-density lipoprotein[7] indicating that this receptor could be involved in the modulation of cholesterol availability. Scavenger receptors have previously been implicated in corticosteroid output from the adrenal gland. Specifically, Scavenger Receptor BI (SR-BI) was shown to regulate glucocorticoid responses via binding of serum cholesterol, the necessary substrate for adrenal corticosteroids [8]–[11]. Aldosterone, the other major adrenal cortex-derived corticosteroid, is known to be regulated by the renin-angiotensin-aldosterone system, which is situated in the adrenals, lung, and kidney.[12]. While a role for scavenger receptors in regulating glucocorticoid responses has been demonstrated, the role of macrophage-expressed scavenger receptors in the regulation of adrenal corticosteroid output at steady state has not yet been explored.

Results

It is known that statins, cholesterol-lowering drugs have been shown to reduce aldosterone levels in humans [13],[14]. Moreover, *in vitro* studies have demonstrated that cholesterol supplementation boosts the production of aldosterone from cultured cells [15]–[17]. Given that the adrenal-derived mouse corticosteroids, most notably corticosterone and aldosterone, derive from cholesterol as the common precursor (**Fig. 1a**) we hypothesised that cholesterol binding scavenger receptors could modulate adrenal corticosteroid output by regulating the availability of cholesterol that could feed into the steroid hormone biosynthetic pathway. To test this hypothesis, we measured the concentrations of aldosterone and corticosterone in the plasma of Marco$^{-/-}$ and wild-type mice. We found that male Marco$^{-/-}$ mice had significantly elevated levels of plasma aldosterone relative to wild-type (WT) mice (**Fig. 1b**). In contrast, plasma corticosterone levels were not significantly altered in male mice lacking Marco (**Fig. 1c**). Marco-deficient female mice did not have altered levels of aldosterone relative to WT, but plasma corticosterone was increased relative to WT counterparts (**Fig 1d, e**). We observed that the adrenal glands from Marco-deficient male mice were significantly lighter than WT controls in male but not female mice (**Fig. 1f, g**). To establish whether cholesterol could explain the elevated plasma aldosterone we observe in Marco-deficient male mice, we measured the levels of total serum cholesterol and intra-adrenal cholesterol in males. We found that Marco$^{-/-}$ mice had reduced serum cholesterol relative to WT controls (**Fig. 1h**), while the normalised levels of intra-adrenal cholesterol were similar between both mouse strains (**Fig. 1i**). Taken collectively, these findings suggest that, while Marco-deficient male mice have elevated plasma aldosterone concentrations, this is not dependent on systemic or intra-adrenal cholesterol availability. For the purposes of this paper we focused on the phenotype evident in male mice, namely the increase in plasma aldosterone concentrations.

We next hypothesised that adrenal gland-derived *Marco* could play a role in modulating aldosterone output. Analysis of publicly available Single cell sequencing data from murine adrenal glands shows that adrenals do contain a substantial population of macrophages expressing *Ptprc* (CD45), *Adgre1* (F4/80), and *Cd68* (CD68). However, we did not detect Marco expression in this cluster of cells, nor any other cluster identified in our analyses (**Fig. 2A**). This finding was corroborated by immunostaining of male murine adrenal glands, which showed CD68^{+} macrophages in the adrenal zona fasciculata and zona glomerulosa that did not stain positively for MARCO (**Fig. 2b**). Given that the lung is another site in the RAAS axis, we postulated that Marco-expressing cells in the lung could be involved in mediating the aldosterone phenotype we observed in **Fig. 1b**. Indeed, single cell RNA seq analysis of the murine lung shows that *Ptprc* (CD45), *Adgre1* (F4/80), and *Cd68* (CD68) expressing cells (Alveolar Macrophages) also express

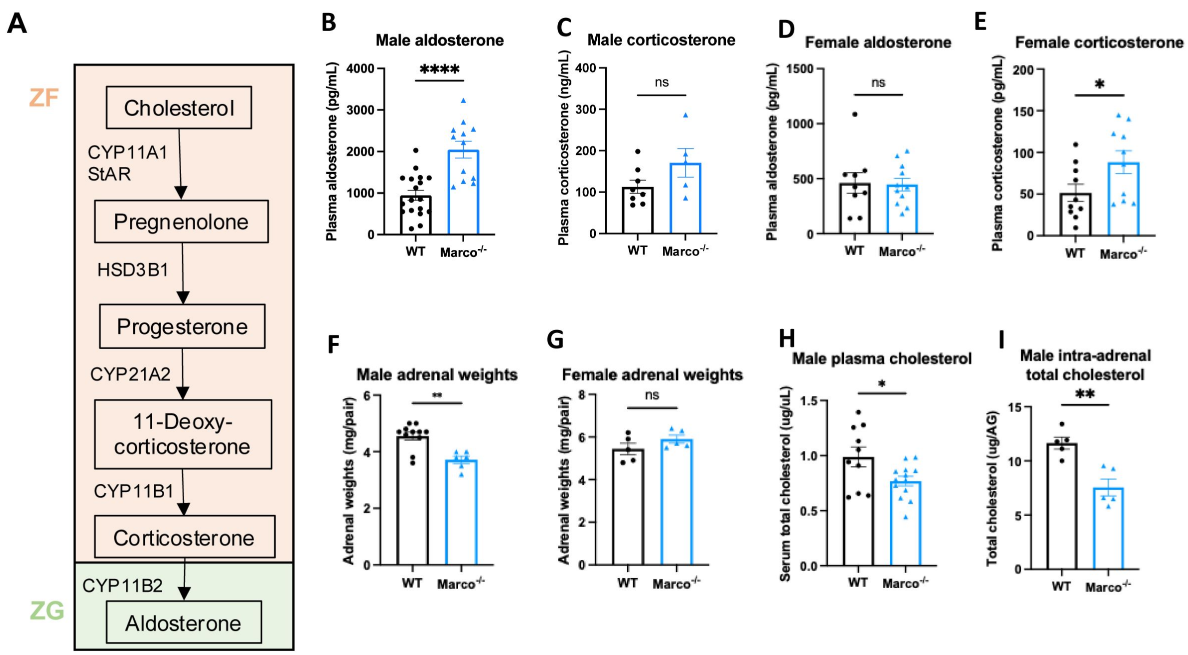

Figure 1.

Marco-deficient mice have elevated aldosterone and reduced serum cholesterol.

(A) Schematic depicting the murine adrenal corticosteroid biosynthetic pathway. Plasma aldosterone and corticosterone concentrations from wild type (WT) and Marco-deficient male (B, C) and female (D, E) mice as measured by ELISA. (F, G) Weights of both adrenal glands from WT or Marco-deficient male and female mice. Plasma total cholesterol levels (H) and relative intra-adrenal cholesterol levels, normalised to adrenal weight (I), in WT and Marco-deficient male mice. Data in B-H were analysed by two-tailed unpaired Student's *t*-test and are shown as average ± s.e.m. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, **** $P < 0.0001$.

*Marco (***Fig 2c**), a finding corroborated by immunostaining in the lung (**Fig. 2c**). To further validate our single cell sequencing and immunofluorescence data, we carried out qPCR for *Marco* in the lungs and adrenal glands from male WT and Marco$^{-/-}$ mice, which further demonstrated that the lung is the primary site of Marco expression in the RAAS (**Fig. 2e**).

Aldosterone is a potent blood pressure-regulating hormone, the dysregulation of which can cause severe hypertension and increased cardiovascular risk. It therefore follows that its production is tightly regulated. Aldosterone biosynthesis is fundamentally regulated intra- adrenally by cytochrome P450 family members in the corticosteroid biosynthetic pathway (**Fig. 1a**). We therefore tested whether altered expression of enzymes in this pathway could explain the hyperaldosteronism observed in Marco-deficient mice. Marco$^{-/-}$ male and female mice showed similar expression of aldosterone biosynthetic enzymes (*Star, Cyp11a1. Hsd3b1, Cyp11b1, Cyp11b2)* as wild type mice (**Fig. 3a, b**). While *Cyp11b2* (aldosterone synthase) is only expressed in the adrenal zona glomerulosa, other biosynthetic enzymes essential for aldosterone production are expressed in the zona fasciculata.

CYP11B1 catalyses the conversion of 11-deoxycorticosterone to corticosterone. Corticosterone can be catalysed to aldosterone by CYP11B2. In this sense, the route via CYP11B1 is a bona fide route for the generation of aldosterone, as evidenced by the fact that *Cyp11b1* deletion in mice results in a significant reduction in aldosterone production [18]. This route is one that is suppressible via the suppressive action of dexamethasone on ACTH and *Cyp11b1* expression. Moreover, ACTH is a known stimulator of aldosterone [19,20]. Dexamethasone-mediated suppression of the zona fasciculata (**Fig. 3c**) [37] was used to test whether the elevated aldosterone phenotype was zona fasciculata-dependent. Marco-deficient mice fed dexamethasone-supplemented drinking water had plasma aldosterone concentrations comparable to vehicle-treated mice (**Fig. 3d**), indicating zona fasciculata activity does not contribute to elevated aldosterone levels in Marco$^{-/-}$ mice. Taken collectively, these findings indicate that elevated aldosterone observed in Marco-deficient mice arises extra-adrenally and can therefore be considered a form of secondary hyperaldosteronism. Kidney-derived renin is the initiating hormone in the enzymatic cascade that generates angiotensin II, a potent stimulator of adrenal aldosterone production. We therefore compared plasma renin activity between male WT and Marco$^{-/-}$ mice, finding no significant differences between the two strains (**Fig. 3e**).

Next, we investigated whether lung-derived angiotensin converting enzyme (*Ace*) could explain the elevated aldosterone levels observed in Marco$^{-/-}$ mice. ACE in the lung catalyses the conversion of Angiotensin I to the aldosterone-stimulating peptide Angiotensin II. We carried out a qPCR test for *Ace* in the lungs of WT and Marco$^{-/-}$ mice in both sexes, finding that Marco-deficient male animals had elevated levels of lung *Ace* relative to WT controls in male mice only (**Fig. 4a**). Immunofluorescent staining of WT and Marco$^{-/-}$ lungs revealed a substantially higher level of ACE protein in Marco-deficient male mice, while myeloid presence, as measured by CD68 staining, remained unchanged (**Fig. 4b**). While low levels of ACE expression could be detected in CD68^{+} cells (data not shown), the vast majority of ACE was outside of monocytes and macrophages. We used image analysis software to quantify these changes, finding that ACE median fluorescence intensity (MFI) was significantly increased in male Marco-deficient lungs, while CD68^{+} myeloid cells were present in wild type and knock- out animals at similar levels (**Fig. 4c, d**). Myeloid cell numbers and lung ACE expression were similar in WT and Marco-deficient female lungs (**Fig. 4e, f**). We also measured the levels of plasma potassium and sodium levels in male and female WT and Marco-deficient animals, but observed no differences between the genotypes (**Fig. 4g-j**). Since aldosterone is a known regulator of blood pressure via regulation of blood fluid balance, we also measured blood pressure using the tail-cuff method. We observed that Marco-deficient male mice had marginally reduced diastolic blood pressures, but systolic and mean blood pressure were no different between the two strains (**Fig. 4k-m**).

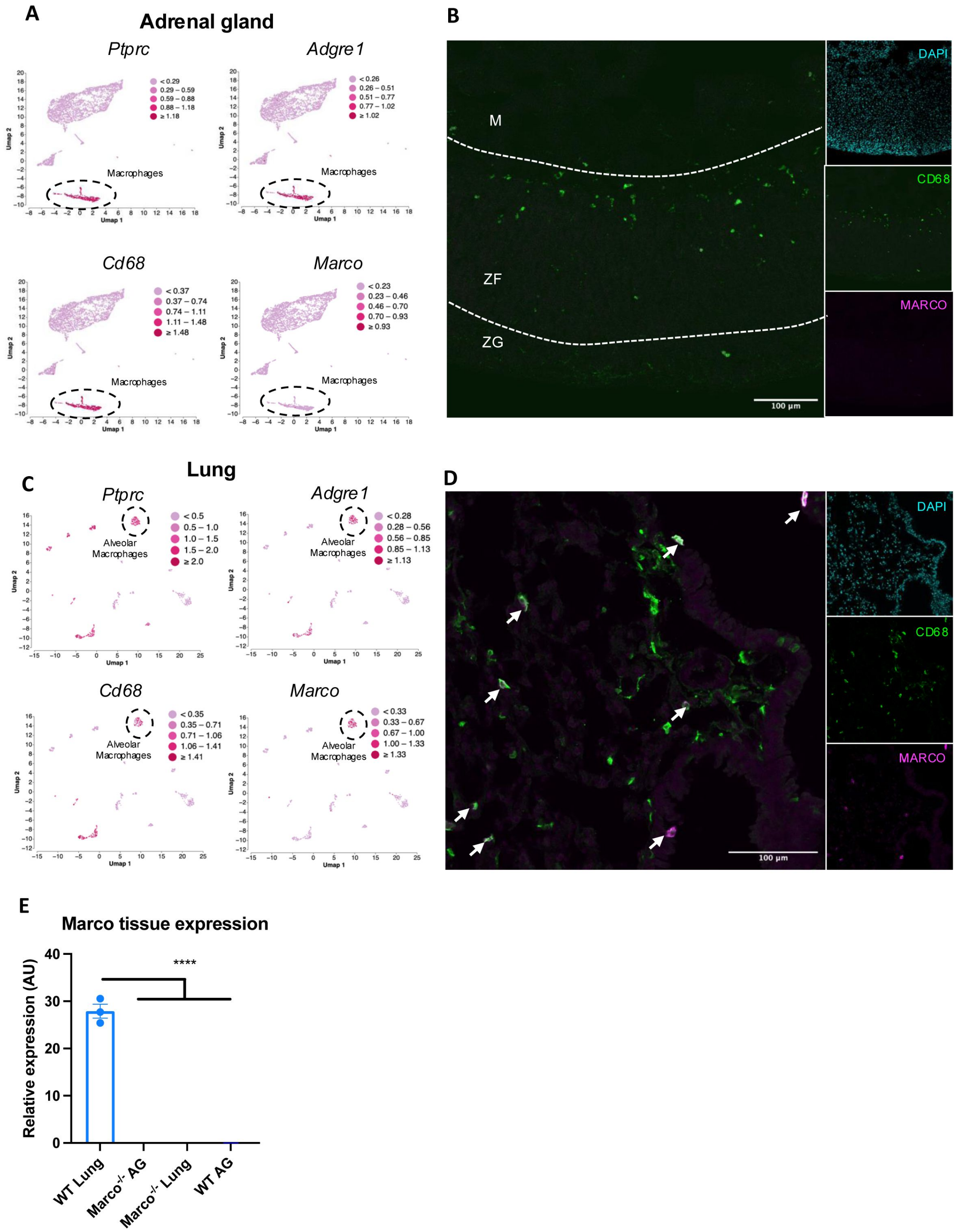

Figure 2.

Marco is expressed in the lung and not adrenal glands.

(A) Single cell RNA seq data plots from PMID 33571131 representing mRNA expression of Ptprc (CD45), Adgre1 (F4/80), Cd68, and Marco, in male murine adrenal glands. (B) Representative image showing a central cryosection of the male murine adrenal gland from a C57bl/6 mouse stained against CD68 (green) Marco (magenta), and DAPI (cyan). (C) Single cell RNA seq data plots from PMID 30283141 representing mRNA expression of Ptprc (CD45), Adgre1 (F4/80), Cd68, and Marco, in the male murine lung. (D) Representative image showing a cryosection of the male murine lung from a hCD68-GFP reporter mouse stained against GFP (green) Marco (magenta), and DAPI (cyan). (E) qPCR data showing the relative gene expression data for *Marco* in the indicated tissues. M = medulla, ZF = zona fasciculata, ZG = zona glomerulosa.

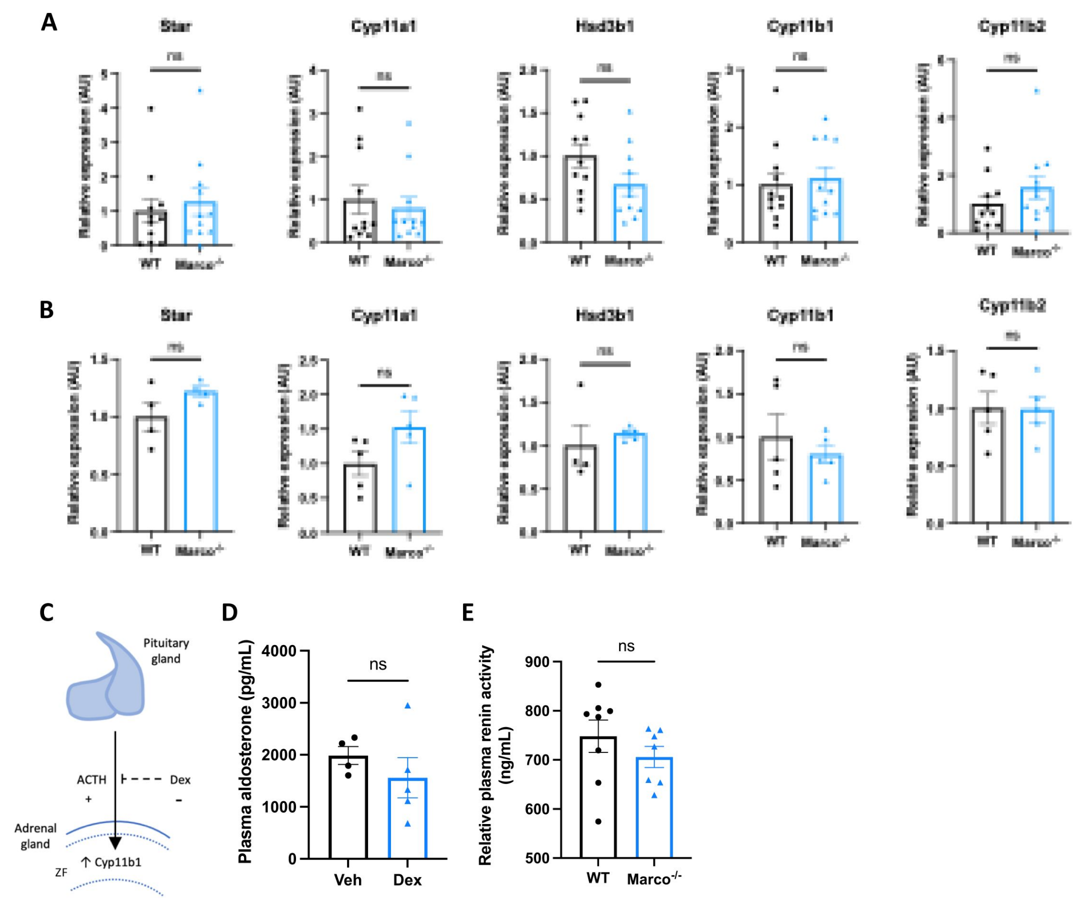

Figure 3.

Marco-mediated elevation of aldosterone is not explained by upregulation of adrenal biosynthetic enzymes, the zona fasciculata, or renin.

qPCR data reporting the expression of adrenal corticosteroid biosynthetic enzymes between wild type (WT) and Marco-deficient male mice in male (A) and female (B) mice. (C) A schematic illustrating that pituitary-derived adrenocorticotropic hormone (ACTH) stimulates the upregulation of Cyp11b1 from the zona fasciculata (ZF) and the dexamethasone-mediated suppression of this effect. (D) A bar graph showing the plasma aldosterone concentrations of Marco-deficient mice treated with vehicle or dexamethasone-supplemented drinking water for 14 days. (E) A bar graph showing the plasma renin activity of wild type (WT) and Marco-deficient mice at steady state. All data were analysed by two-tailed unpaired Student's *t*-test and are shown as average ± s.e.m. ns P>0.05. * $P < 0.05$.

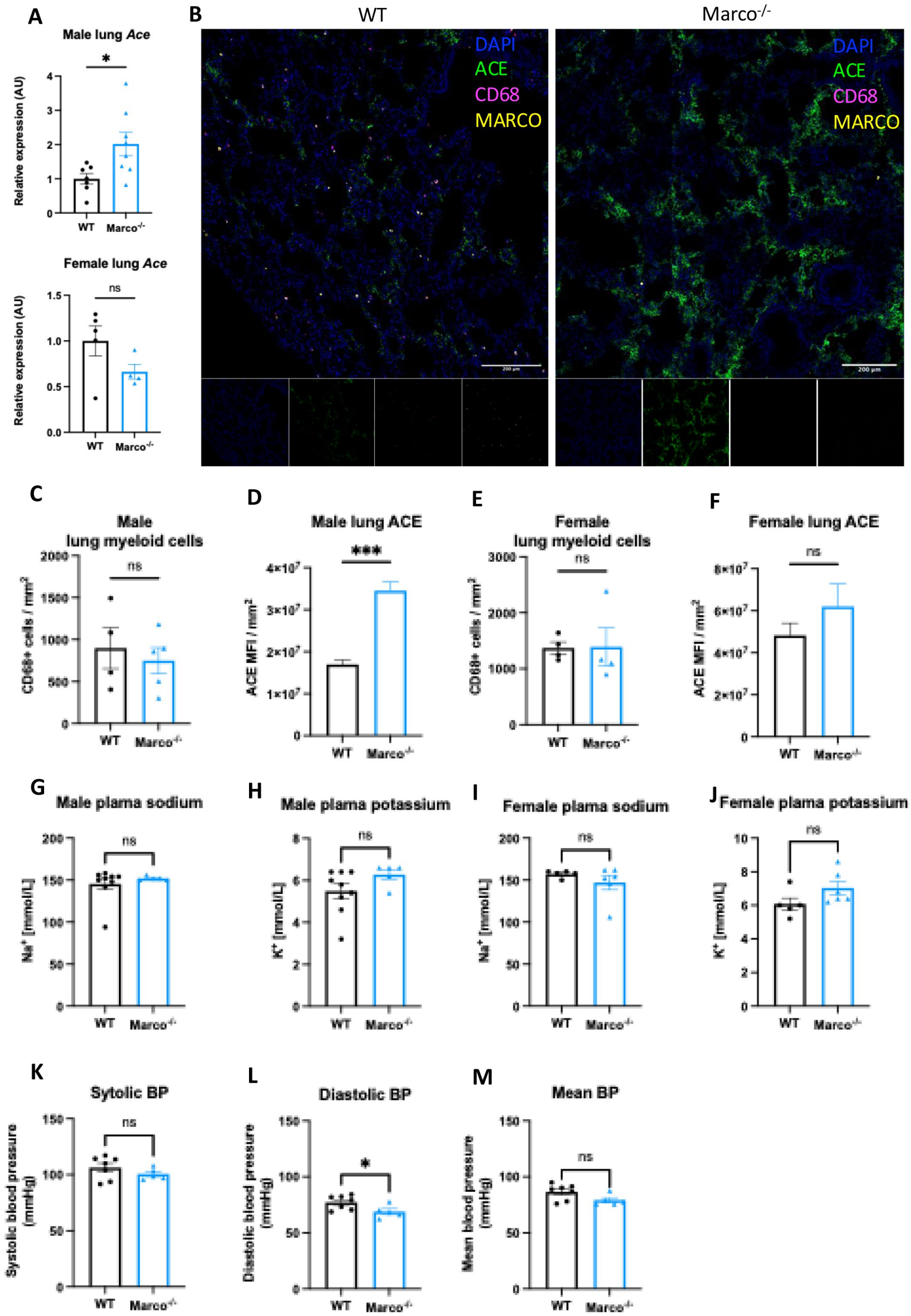

Figure 4.

Marco-deficient mice have enhanced expression of Ace in the lung.

(A) qPCR data reporting the expression of angiotensin converting enzyme (*Ace*) in the lungs of wild type (WT) and Marco-deficient male and female mice. (B) Representative images showing a cryosection of wild-type and Marco-deficient male murine lungs stained against ACE (green) CD68 (magenta), MARCO (yellow) and DAPI (blue). Quantitation of DAPI-normalised ACE median fluorescence intensity (MFI) (C) and $CD68^{+}$ myeloid cell presence in the lungs of wild type and Marco-deficient male and female mice (C-F). Plasma sodium and potassium levels in male and female mice (G-J). Systolic, diastolic, and mean blood pressure measurements in male mice of the indicated genotypes (K-M). Data in A, C-M, were analysed by two-tailed unpaired Student's *t*-test and are shown as average ± s.e.m. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, **** $P < 0.0001$.

We then turned back to analysis of single cell RNA-seq data to identify the cells in the lung that could be mediating this effect. In the lung, unsupervised cluster analysis revealed a total of 12 cell clusters with distinct gene expression signatures (**Fig. 5a**). We determined the identity of each cell cluster based on the expression of established cell type-specific marker genes, aided by the marker genes identified and outlined in **Fig. 5b**. We next used dot plots to visualize expression of *Marco* and *Ace* across the different cell clusters. We observed notable *Marco* expression only in alveolar macrophages amongst the different cell clusters (**Fig. 5c**). *Ace* was shown to be primarily expressed by lung endothelial clusters 1 and 2 (**Fig. 5c**), in agreement with what is known about lung *Ace* expression. To test whether alveolar macrophages are capable of suppressing endothelial cell *Ace* expression, we co-cultured the MPI alveolar macrophage cell line [21], with or without deletion of Marco, with HUVECs endothelial cells for 24 hours, and measured *Ace* expression via qPCR. We observed an increase in *Ace* expression in the Marco-deficient MPI co-cocultures but not WT, though this did not reach statistical significance. Taken collectively, these data suggest a model whereby Marco-expressing alveolar macrophages may, in responses to an as-of-yet unidentified factor, inhibit *Ace* expression at the gene and protein level, and thereby negatively regulate the cleavage of Angiotensin I to form Angiotensin II, and thereby aldosterone production (**Fig. 5e**).

In conclusion, we hereby demonstrate that Marco is a negative regulator of aldosterone production, associated with a suppression of angiotensin converting enzyme expression in the lungs of male mice. We propose a model in which extra-adrenal Marco expressing alveolar macrophages, through tissue crosstalk with lung endothelial cells, actively inhibit Ace expression and thereby inhibit the production of aldosterone from the adrenal glands.

Discussion

Multiple studies over preceding decades have shown that macrophages are much more than simply immune and inflammatory cells. Macrophages have been shown to mediate a wide variety of physiological functions and support homeostasis in virtually every tissue they are found [22–27]. Our original hypothesis centred on the regulation of corticosteroid output via modulation of cholesterol availability. While we ended up disproving this hypothesis, we did provide a novel example of the non-immune functions of tissue macrophages. It was recently shown that tissue infiltrating macrophages cause neuronal cell death and effective denervation, which disrupts normal melatonin secretion in a model of cardiovascular disease [28]. To our knowledge, ours is the first study to report that macrophages are involved in the regulation of hormonal output *in vivo* at steady state. It is known that immunosuppressants such as cyclosporine can increase blood pressure [29] yet the mechanism remains incompletely understood. This observation is consistent with our data showing macrophage-mediated negative regulation of the blood pressure-controlling hormone aldosterone. Additionally, the fact that higher ACE/ACE2 ratios have been linked to hypertension and worse outcomes in Covid-19 [30] could implicate MARCO in the susceptibility to severe disease.

Secondary aldosteronism (SA) is distinguished from primary aldosteronism through renin – SA is characterised by high aldosterone and high or non-suppressed renin levels [31]. By this definition, Marco-deficient mice exhibit SA, due to their high aldosterone and normal renin levels. SA is also defined as elevated aldosterone that originates outside of the adrenal gland – in this case from ACE in the lung. This finding raises the intriguing possibility that aldosteronism could be triggered or aggravated by MARCO-binding ligands. The fact that blood pressure and plasma electrolytes are not substantially modulated by Marco deletion, despite elevation in plasma aldosterone concentrations, indicate that this mechanism has not evolved to regulate systemic physiology. However, ACE and Angiotensin II are pleiotropic molecules. In addition to their physiological roles, they are also exhibit immunomodulatory functions. *In vitro*, lipopolysaccharide (LPS) treatment of the HUVECs endothelial cell line or human primary endothelial cells causes a

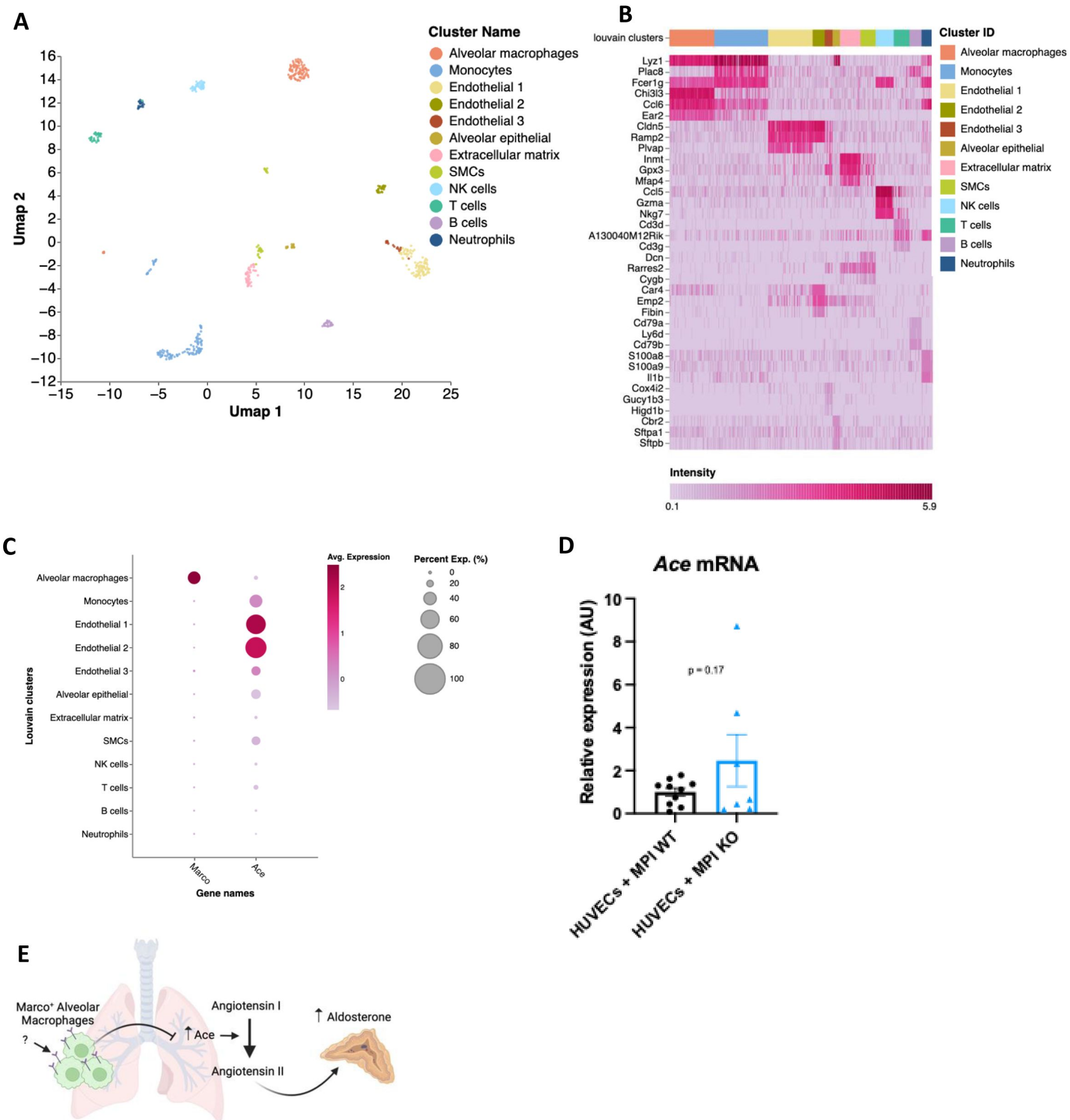

Figure 5.

A proposed model for macrophage-mediated regulation of lung Ace expression.

(A) Single cell RNA seq UMAP plot depicting the cell types present in the male murine lung. (B) Marker heatmap showing the top three gene markers for each cell cluster in (A). (C) A dot plot showing the gene expression levels of Marco and Ace in the different cell clusters of the male murine lung. (D) qPCR data showing the relative expression levels of Ace in co-coltures contraining HUVECs cells with Marco sufficient or deficient MPI macrophages. (E) A schematic, generated using *BioRender.com/fhfe2nn*, showing the working model by which Marco+ alveolar macrophages regulate aldosterone output from the adrenal gland. Data in D were analysed by two-tailed unpaired Student's *t*-test and are shown as average ± s.e.m. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, **** *P* < 0.0001.

reduction in ACE activity, mRNA and protein levels. Similarly, ACE expression in human lung tissue is reduced in sepsis patients relative to control tissues [32,33]. Supernatant from LPS or TNFα-treated endothelial cells shows a high proportion of ACE^+ endothelial microparticles (EMPs). ACE^+ EMPs are also increased in plasma of mice undergoing experimental lung injury and in human septic patients [33], indicating shedding of lung endothelial ACE in the context of inflammation. Ace and AngII have been shown to enhance inflammation in a variety of contexts. It may therefore be that Marco-expressing alveolar macrophages exert an immune-modulatory function within the murine lung via controlling tissue *Ace* expression. Alveolar macrophages, which use MARCO to bind and uptake bacteria [34], could serve to enhance lung immunity in a pathogen and ACE-dependent manner. Pharmacologic ACE inhibition was previously shown to dampen pro-inflammatory immune cell inflammation in radiation-induced pneumonitis [35]. Future experiments could therefore explore whether Marco regulates lung tissue Ace expression in the context of inflammation and/or pathogenic challenge. While acute increases of plasma aldosterone have varying effects on endothelial function in humans and animal models, chronic (i.e. genetic) elevations of plasma aldosterone are virtually universally associated with endothelial dysfunction [36]. This is frequently characterised by reduced endothelial NOS activity from, and acetylcholine action on, endothelial cells [36]. It would therefore be logical to test whether vascular dysfunction is present in the endothelia of Marco-deficient mice relative to wild type mice.

While the data presented here indicate that Marco deletion is a positive regulator of aldosterone and lung Ace expression, important questions remain to be answered, such as the precise mechanism by which Marco-expressing alveolar macrophages suppress Ace expression, and the ligand(s) responsible for stimulating this effect via the receptor. Additionally, the mechanisms mediating the sex differences observed here remain unexplored. Our findings nevertheless introduce a novel immune paradigm the renin angiotensin aldosterone system.

Methods

Mice

C57BL/6J, $Marco^{-/-}$, and $hCD68^{GFP/GFP}$ mice were bred and housed in individually ventilated cages in specific pathogen-free conditions at the University of Oxford. All experiments on mice were conducted according to institutional, national, and European animal regulations. Animal protocols were approved by the animal welfare and ethics review board of the Department of Physiology, Anatomy, and Genetics at the University of Oxford.

Tissue collection

All tissues were collected from 8-11 week old male mice that were culled between 0930-1100. Mice were euthanised by intraperitoneal injection of 10μL/g 200mg/mL Pentoject (pentobarbital sodium; Animalcare). Following cessation of the pedal motor reflex a sample of blood was taken from the right atrium prior to perfusion through the left ventricle with 20mL 1X phosphate-buffered saline (PBS; Sigma Aldrich). Adrenal glands were dissected, and the peri-adrenal fat removed using a stereomicroscope. Following harvest, adrenal glands and lungs were frozen at -80°C until subsequent molecular analyses or fixed overnight in 4% paraformaldehyde (PFA) prior to histological analyses. Bloods obtained prior to perfusion were collected into EDTA coated tubes to prevent clotting and centrifuged at 1000g for 15 minutes to separate plasma. Plasma samples were frozen at -80°C in 1.5mL Eppendorf tubes until subsequent analyses.

Plasma hormone, cholesterol, renin activity, and electrolyte quantitation

We measured plasma aldosterone and corticosterone using reverse ELISA kits (Enzo Life Sciences) according to manufacturer's instructions. Plasma cholesterol was quantified by colorimetric assay (Cholesterol Quantitation Kit; Sigma Aldrich) according to manufacturer's instructions. Plasma renin activity was measured by fluorometric assay using the Renin Assay Kit (Sigma Aldrich). Colorimetric and fluorimetric readouts from these assays were attained using a FLUOstar Omega plate reader (BMG Labtech). Plasma electrolytes were outsourced to Pinmoore Animal Laboratory Services Limited (Cheshire, England) and sodium and potassium electrolyte readings were taken using an I-Smart 30 Vet electrolyte analyser (Woodley Equipment).

Single cell sequencing data analysis

Publicly available single cell RNA-seq datasets (steady-state adrenal: PMID 33571131, GEO accession number GSE161751; steady-state lung: PMID 30283141, GEO accession number GSE109774) were processed, explored and visualised using Cellenics® community instance (*https://scp.biomage.net/*) hosted by Biomage (*https://biomage.net/*). For the adrenal dataset the classifier filter was disabled as the sample had been pre-filtered. The cell size distribution filter was disabled. A mitochondrial content filter was used, with the absolute threshold method used (bin step = 0.05, maximum fraction = 0.1). The number of genes vs UMI filter was used with a linear fit and P value = 0.0004906771. The doublet filter was used (bin step = 0.05, probability threshold = 0. 8743427. The harmony algorithm was used for data integration (2000 genes, log normalisation). The RPCA method was used for dimensionality reduction (13 principal components). Cells and clusters were visualised using UMAP embedding with the cosine distance metric (minimum distance = 0.2). Louvain clustering was the clustering algorithm used in this analysis (resolution = 0.6). For the lung dataset the classifier filter was disabled as the sample had been pre-filtered. A cell size distribution filter was utilised with binStep = 200 and minimum cell size = 1669. A mitochondrial content filter was used, with the absolute threshold method used, bin step = 0.3, maximum fraction = 0. The number of genes vs UMI filter was used with a linear fit and P value = 0.001. The doublet filter was used with bin step = 0.02 and probability threshold = 0.7616616. The harmony algorithm was used for data integration (2000 genes, log normalisation). The RPCA method was used for dimensionality reduction (24 principal components). Cells and clusters were visualised using UMAP embedding with the cosine distance metric (minimum distance = 0.3). Louvain clustering was the clustering algorithm used in this analysis (resolution = 0.8).

Cryo-sectioning and antibody staining

Prior to tissue embedding, adrenal glands were cryoprotected in 30% sucrose solution (in PBS) overnight. Cryoprotected adrenal glands were embedded in OCT embedding medium (Thermo Fisher Scientific) and snap frozen in liquid nitrogen. 15um sections were cut from embedded tissues using a Leica cryostat and mounted onto charged microscope slides. Cryosectioned tissue slices were thawed in PBS for 10 min at room temperature, tissue sections were circled using a super PAP pen (Life Technologies), blocked and permeabilised for 1 hour at room temperature in perm/block solution (3% bovine serum albumin, 2% goat serum, 1% Triton X-100, 0.01% NaN3 in PBS). Slides were stained with CD68-AF647 (clone FA-11, BioLegend), chicken anti-GFP (ab13970, Abcam) with goat anti-chicken AF488 (A11039, Invitrogen), mouse anti-MARCO (ED31, BioRad) with goat anti-mouse AF546 (A-11030, Invitrogen), and rabbit anti-ACE (MA5-32741, Invitrogen) with goat anti-rabbit Alexa Fluor 488 Tyramide Super Boost kit (Invitrogen, B40922). Primary stains were done overnight at 4°C and secondary stains were done for 1 hour at room temperature. The goat anti-rabbit Alexa Fluor 488 Tyramide Super Boost kit was used as per manufacturer's instructions. DAPI counterstains were done at 1:1000 for 5 minutes at room temperature. Fluoromount G mounting medium (Invitrogen) was used to mount coverslips to slides which were then sealed with clear nail varnish. Immunofluorescence images were acquired using the Zeiss 880 confocal microscope and images were analysed in FIJI.

RT-PCR

Tissues were homogenised using the Precellys Hard Tissue Grinding Kit (MK28-R; Bertin Technologies). Total RNA from homogenised adrenals or lungs was isolated using RNeasy Plus Micro Kit (Qiagen, cat# 74034). cDNA was reverse transcribed using SuperScript II (Invitrogen) and random primers (Invitrogen). Quantitative PCR was performed using SYBR Green (Applied Biosystems) in C1000 Touch™ Thermal Cycler (BioRad). B-actin was used as the housekeeping gene to normalize samples. We used the following formula to calculate the relative expression levels: $RQ = 2^{\wedge -\Delta Ct} \times 100 = 2^{-(\text{Ct gene of interest} - \text{Ct } \beta \text{ actin})} \times 100$. The following mouse-specific primers were used. *Star* forward 5’-CAGGGCCAAGAAAACCTACA-3’; *Star* reverse 5’-ACGAGCATTTTGAAGCACCT-3’; *Cyp11a1* forward 5’-AGGACTTTCCCTGCGCT -3’; *Cyp11a1* reverse 5’-GCATCTCGGTAATGTTGG-3’; *Hsd3b1* forward 5’-GCGGCTGCTGCACAGGAATAAAG-3’; *Hsd3b1* reverse 5’-TCACCAGGCAGCTCCATCCA-3’; *Cyp11b1* forward 5’-TCACCATGTGCTGAAATCCTTCCA-3’; *Cyp11b1* reverse 5’-GGAAGAGAAGAGAGGGCAATGTGT-3’; *Cyp11b2* forward 5’-CAGGGCCAAGAAAACCTACA-3’; *Cyp11b2* reverse 5’-ACGAGCATTTTGAAGCACCT-3’; *B-actin* forward 5’-TCATGAAGTGTACGTGGACATCC-3’; *Ace* forward 5’-GCTTCCTCTTTCTGCTGCTCTG -3’; *Ace* reverse 5’-TGCCCTCTATGGTAATGTTGGT- 3’; B-*actin* reverse 5’-CCTAGAAGCATTTGCGGTGGACGATG-3’.

Zona fasciculata suppression model

Water-soluble dexamethasone (Sigma-Aldrich) was reconstituted in autoclaved deionized water at the concentration of 0.0167 mg/mL as described in [37]. 10 week old Marco$^{-/-}$ mice were fed dexamethasone-supplemented drinking water for 14 days prior to blood sampling as described above.

Digital image analysis

Quantitation of the area and intensities of immunofluorescent stains was done using FIJI image analysis software. Three cryosections of stained lung tissue (~5mm^2 tissue acquired per section) were analysed per mouse. CD68$^+$ cells were segmented using the Otsu thresholding plugin. The parameters were adjusted manually to ensure optimum cell segmentation. DAPI-stained nuclei were segmented using the Otsu thresholding plugin. Median fluorescence intensity (MFI) for ACE was measured via the integrated density of the stain, which was normalised to the area of DAPI staining.

Blood pressure readings

Systolic, diastolic, and mean blood pressures were taken using the CODA high throughput noninvasive blood pressure system (Kent Scientific). Mice were acclimatised to animal holders on three occasions in the week prior to experimental readings.

Co-culture

X63-GMCSF and WT and Marco-deficient MPI cells were a kind gift from Dr Subhankar Mukhopadhyay. 20,000 HUVECs cells were plated with 20,000 WT or Marco-deficient MPI cells in 96 well plates for 24 hours in high glucose DMEM, 10% Fetal Bovine Serum (FBS), 1% X63-GM-CSF conditioned media as a source of GM-CSF, 1% Penicillin/Streptomycin (P/S), and 1X MesoEndo Cell growth medium (all bought from Sigma Aldrich) and incubated at 37°C, 5% CO_2. RNA was extracted and qPCRs performed as previously described.

Statistical Analysis

Results are expressed as the mean ± s.e.m., as indicated in the figure legends. Statistical significance between two experimental groups was assessed using two-tailed student's t test. All Statistical analyses were performed in GraphPad Prism 9 (GraphPad, USA) for Mac OS X. All calculated P values are reported in the figures, denoted by $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, ns $P>0.05$.

Additional information

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Reviewer #2 (Public review):

Summary:

Tissue-resident macrophages are more and more thought to exert key homeostatic functions and contribute to physiological responses. In the report of O'Brien and Colleagues, the idea that the macrophage-expressed scavenger receptor MARCO could regulate adrenal corticosteroid output at steady-state was explored. The authors found that male MARCO-deficient mice exhibited higher plasma aldosterone levels and higher lung ACE expression as compared to wild-type mice, while the availability of cholesterol and the machinery required to produce aldosterone in the adrenal gland were not affected by MARCO deficiency. The authors take these data to conclude that MARCO in alveolar macrophages can negatively regulate ACE expression and aldosterone production at steady-state and that MARCO-deficient mice suffer from a secondary hyperaldosteronism.

Strengths:

If properly demonstrated and validated, the fact that tissue-resident macrophages can exert physiological functions and influence endocrine systems would be highly significant and could be amenable to novel therapies.

Major weakness:

The comparison between C57BL/6J wild-type mice and knock-out mice for which a precise information about the genetic background and the history of breedings and crossings is lacking can lead to misinterpretations of the results obtained. Hence, MARCO-deficient mice should be compared with true littermate controls.

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Author response:

The following is the authors' response to the original reviews

We again thank the reviewers for their comments and recommendations. In response to the reviewer's suggestions, we have performed several additional experiments, added additional discussion, and updated our conclusions to reflect the additional work. Specifically, we have performed additional analyses in female WT and Marco-deficient animals, demonstrating that the Marco-associated phonotypes observed in male mice (reduced adrenal weight, increased lung Ace mRNA and protein expression, unchanged expression of adrenal corticosteroid biosynthetic enzymes) are not present in female mice. We also report new data on the physiological consequences of increased aldosterone levels observed in male mice, namely plasma sodium and potassium titres, and blood pressure alterations in WT vs Marco-deficient male mice. In an attempt to address the reviewer's comments relating to our proposed mechanism on the regulation of lung Ace expression, we additionally performed a co-culture experiment using an alveolar macrophage cell line and an endothelial cell line. In light of the additional evidence presented herein, we have updated our conclusions from this study and changed the title of our work to acknowledge that the mechanism underlying the reported phenotype remains incompletely understood. Specific responses to reviewers can be seen below.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *The investigators sought to determine whether Marco regulates the levels of aldosterone by limiting uptake of its parent molecule cholesterol in the adrenal gland. Instead, they identify an unexpected role for Marco on alveolar macrophages in lowering the levels of angiotensin-converting enzyme in the lung. This suggests an unexpected role of alveolar macrophages and lung ACE in the production of aldosterone.*
>
> *Strengths:*
>
> *The investigators suggest an unexpected role for ACE in the lung in the regulation of systemic aldosterone levels.*
>
> *The investigators suggest important sex-related differences in the regulation of aldosterone by alveolar macrophages and ACE in the lung.*
>
> *Studies to exclude a role for Marco in the adrenal gland are strong, suggesting an extra-adrenal source for the excess Marco observed in male Marco knockout mice.*
>
> *Weaknesses:*
>
> *While the investigators have identified important sex differences in the regulation of extrapulmonary ACE in the regulation of aldosterone levels, the mechanisms underlying these differences are not explored.*
>
> *The physiologic impact of the increased aldosterone levels observed in Marco -/- male mice on blood pressure or response to injury is not clear.*
>
> *The intracellular signaling mechanism linking lung macrophage levels with the expression of ACE in the lung is not supported by direct evidence.*

Reviewer #2 (Public Review):

Summary:

Tissue-resident macrophages are more and more thought to exert key homeostatic functions and contribute to physiological responses. In the report of O'Brien and Colleagues, the idea that the macrophage-expressed scavenger receptor MARCO could regulate adrenal corticosteroid output at steady-state was explored. The authors found that male MARCO-deficient mice exhibited higher plasma aldosterone levels and higher lung ACE expression as compared to wild-type mice, while the availability of cholesterol and the machinery required to produce aldosterone in the adrenal gland were not affected by MARCO deficiency. The authors take these data to conclude that MARCO in alveolar macrophages can negatively regulate ACE expression and aldosterone production at steady-state and that MARCO-deficient mice suffer from secondary hyperaldosteronism.

Strengths:

If properly demonstrated and validated, the fact that tissue-resident macrophages can exert physiological functions and influence endocrine systems would be highly significant and could be amenable to novel therapies.

Weaknesses:

The data provided by the authors currently do not support the major claim of the authors that alveolar macrophages, via MARCO, are involved in the regulation of a hormonal output in vivo at steady-state. At this point, there are two interesting but descriptive observations in male, but not female, MARCO-deficient animals, and overall, the study lacks key controls and validation experiments, as detailed below.

Major weaknesses:

(1) According to the reviewer's own experience, the comparison between C57BL/6J wild-type mice and knock-out mice for which precise information about the genetic background and the history of breedings and crossings is lacking, can lead to misinterpretations of the results obtained. Hence, MARCO-deficient mice should be compared with true littermate controls.

(2) The use of mice globally deficient for MARCO combined with the fact that alveolar macrophages produce high levels of MARCO is not sufficient to prove that the phenotype observed is linked to alveolar macrophage-expressed MARCO (see below for suggestions of experiments).

(3) If the hypothesis of the authors is correct, then additional read-outs could be performed to reinforce their claims: levels of Angiotensin I would be lower in MARCO-deficient mice, levels of Antiotensin II would be higher in MARCO-deficient mice, Arterial blood pressure would be higher in MARCO-deficient mice, natremia would be higher in MARCO-deficient mice, while kaliemia would be lower in MARCO-deficient mice. In addition, co-culture experiments between MARCO-sufficient or deficient alveolar macrophages and lung endothelial cells, combined with the assessment of ACE expression, would allow the authors to evaluate whether the AM-expressed MARCO can directly regulate ACE expression.

Recommendations for the authors:

> ***Reviewer #1 (Recommendations For The Authors):***
>
> *(1) Corticosterone levels in male Marco -/- mice are not significantly different, but there is (by eye) substantially more variability in the knockout compared to the wild type. A power analysis should be performed to determine the number of mice needed to detect a similar % difference in corticosterone to the difference observed in aldosterone between male Marco knockout and wild-type mice. If necessary the experiments should be repeated with an adequately powered cohort.*

Using a power calculator (www.gigacalculator.com) it was determined that our sample size of 13 was one less than sufficient to detect a similar % difference in corticosterone as was detected in corticosterone. We regret that we unable to perform additional measurements as the author suggested in the available timeframe.

> *(2) All of the data throughout the MS (particularly data in the lung) should be presented in male and female mice. For example, the induction of ACE in the lungs of Marco-/- female mice should be absent. Similar concerns relate to the dexamethasone suppression studies. Also would be useful if the single cell data could be examined by sex--should be possible even post hoc using Xist etc.*

Given the limitations outlined in our previous response to reviewers it was not possible to repeat every experiment from the original manuscript. We were able to measure the expression of lung Ace mRNA, ACE protein, adrenal weights, adrenal expression of steroid biosynthetic enzymes, presence of myeloid cells, and levels of serum electrolytes in female animals. These are presented in figures 1G, 3B, 4A, 4E, 4F, 4I, and 4J. We have elected to not present single cell seq data according to sex as it did not indicate substantial differences between males and females in Marco or Ace expression and so does not substantively change our approach.

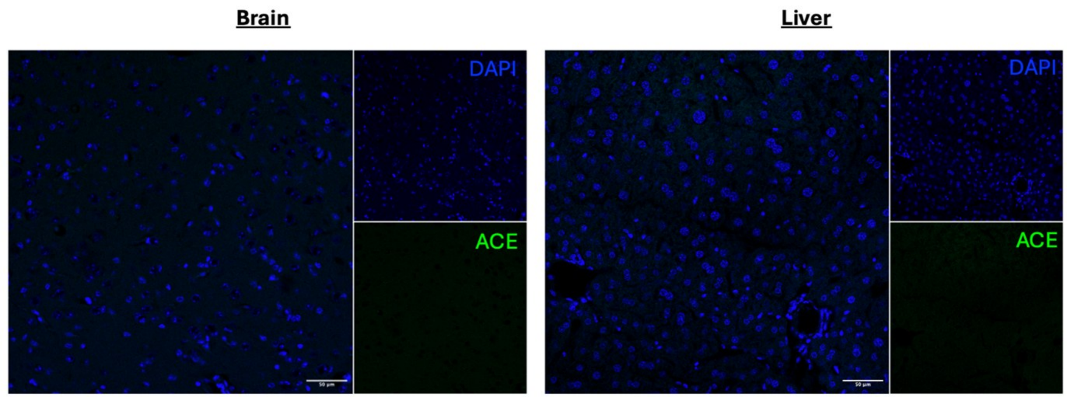
> *(3) IF is notoriously unreliable in the lung, which has high levels of autofluorescence. This is the only method used to show ACE levels are increased in the absence of Marco. Orthogonal methods (e.g. immunoblots of flow-sorted cells, or ideally CITE-seq that includes both male and female mice) should be used.*

We used negative controls to guide our settings during acquisition of immunofluorescent images. Additionally, we also used qPCR to show an increase in Ace mRNA expression in the lung in addition to the protein level. This data was presented in the original manuscript and is further bolstered by our additional presentation of expression data for Ace mRNA and protein in female animals in this revised manuscript.

> *(4) Given the central importance of ACE staining to the conclusions, validation of the antibody should be included in the supplement.*

We don't have ACE-deficient mice so cannot do KO validation of the antibody. We did perform secondary stain controls which confirmed the signal observed is primary antibody-derived. Moreover, we specifically chose an anti-ACE antibody (Invitrogen catalogue # MA5-32741) that has undergone advanced verification with the manufacturer. We additionally tested the antibody in the brain and liver and observed no significant levels of staining.

Author response image 1.



> *(5) The link between alveolar macrophage Marco and ACE is poorly explored.*

We carried out a co-culture experiments of alveolar macrophages and endothelial cells and measure ACE/Ace expression as a consequence. This is presented in figure 5D and the discussion.

> *(6) Mechanisms explaining the substantial sex difference in the primary outcome are not explored.*

This is outside the scope if this project, though we would consider exploring such experiments in future studies.

> *(7) Are there physiologic consequences either in homeostasis or under stress to the increased aldosterone (or lung ACE levels) observed in Marco-/- male mice?*

We measured blood electrolytes and blood pressure in Marco-deficient and Marco-sufficient mice. The results from these experiments are presented in 4G-4M.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Below is a suggestion of important control or validation experiments to be performed in order to support the authors' claims.*
>
> *(1) It is imperative to validate that the phenotype observed in MARCO-deficient mice is indeed caused by the deficiency in MARCO. To this end, littermate mice issued from the crossing between heterozygous MARCO +/- mice should be compared to each other. C57BL/6J mice can first be crossed with MARCO-deficient mice in F0, and F1 heterozygous MARCO +/- mice should be crossed together to produce F2 MARCO +/+, MARCO +/- and MARCO -/- littermate mice that can be used for experiments.*

We thank the reviewer for their comments. We recognise the concern of the reviewer but due to limited experimenter availability we are unable to undertake such a breeding programme to address this particular concern.

> *(2) The use of mice in which AM, but not other cells, lack MARCO expression would demonstrate that the effect is indeed linked to AM. To this end, AM-deficient Csf2rb-deficient mice could be adoptively transferred with MARCO-deficient AM. In addition, the phenotype of MARCO-deficient mice should be restored by the adoptive transfer of wild-*

> *type, MARCO-expressing AM. Alternatively, bone marrow chimeras in which only the hematopoietic compartment is deficient in MARCO would be another option, albeit less specific for AM.*

We recognise the concern of the reviewer. We carried out a co-culture experiments of alveolar macrophages and endothelial cells and measure ACE/Ace expression as a consequence. This is presented in figure 5D and the implications explored in the discussion.

> *(3) If the hypothesis of the authors is correct, then additional read-outs could be performed to reinforce their claims: levels of Angiotensin I would be lower in MARCO-deficient mice, levels of Antiotensin II would be higher in MARCO-deficient mice, Arterial blood pressure would be higher in MARCO-deficient mice, natremia would be higher in MARCO-deficient mice, while kaliemia would be lower in MARCO-deficient mice. Similar read-outs could also be performed in the models proposed in point 2).*

We measured blood electrolytes and blood pressure in Marco-deficient and Marco-sufficient mice. The results from these experiments are presented in 4G-4M.

> *(4) Co-culture experiments between MARCO-sufficient or deficient alveolar macrophages and lung endothelial cells, combined with the assessment of ACE expression, would allow the authors to evaluate whether the AM-expressed MARCO can directly regulate ACE expression.*

To address this concern we carried out a co-culture experiment as described above.

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